

Lecture 6 – Distributions I



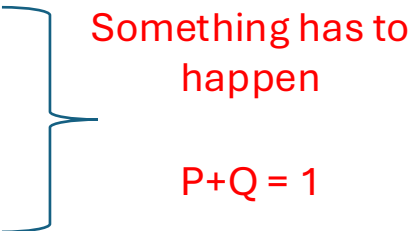
Today's Learning Outcomes

1. Be able to explain the binomial and normal distributions
2. Know pmf/pdf , cdf , ppf , and rvs are and do
3. Know how to test for normality of a dataset

Probability Distributions

- Probability distributions are mathematical functions that describe the probability of an outcome
 - Used to represent how we think processes work and then test our expectations
- Common probability distributions are...
 - Binomial (1 parameter: probability of success)
 - Normal/Gaussian (2 parameters: mean, standard deviation)
 - Uniform (2 parameters: minimum, maximum)
- Most distributions can be described using just a handful of parameters

Binomial Models (discrete events)

- A model of events which act like coin flips
 - heads/tails, present/absent, success/failure
 - Single yes/no outcomes (discrete events)
 - P = probability of a positive outcome
 - $Q = 1 - P$ = probability of a negative outcome
 - For a fair coin, $P = 1/2$, $Q = 1/2$
- 
- Something has to happen
- $P + Q = 1$

Flip 1

heads

tails



If we flip a coin
twice there are
only 4 possible
outcomes

Flip 2

heads

tails



Flip 2

heads

tails



For a series of events, if the events are independent, then the probability of a particular sequence is the product of the probability of the events

Outcome of coin flips	Probability
HH	$0.5 * 0.5 = 0.25$
HT	$0.5 * 0.5 = 0.25$
TH	$0.5 * 0.5 = 0.25$
TT	$0.5 * 0.5 = 0.25$

For a series of events, if the events are independent, then the probability of **at least** X number of “successes” is additive

Probability of...	Sequence	Probability
Two heads	HH	0.25
One head	HT and TH	$0.25 + 0.25 = 0.5$
No heads	TT	0.25

Permutations & Combinations

Combination = a set where order does **not** matter ($1,2,3,3 = 2,1,3,3 = 3,1,2,3$)

Permutation = a set where order **does** matter ($1,2,3,3 \neq 2,1,3,3 \neq 3,1,2,3$)

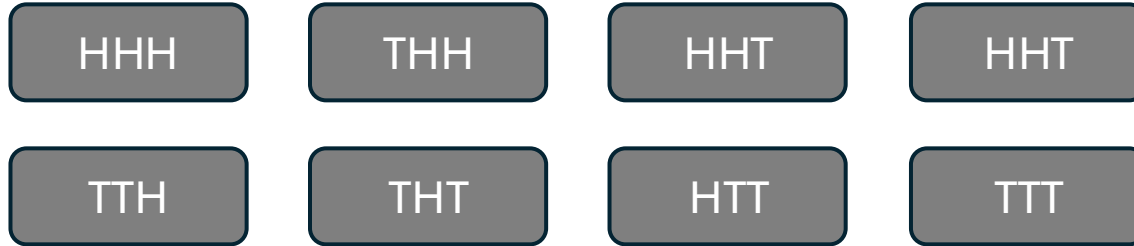
In two coin flips we have four outcomes, with equal probabilities, but two of them correspond to one head HT, TH

There are two permutations (orderings) of the events that produces the combination of one head and one tail

Probability of...	Sequence	Permutations	Probability
Two heads	HH	1	0.25
One head	HT and TH	2	$0.25 + 0.25 = 0.5$
No heads	TT	1	0.25

What if we flip coin thrice?

- How many possible outcomes do we have?



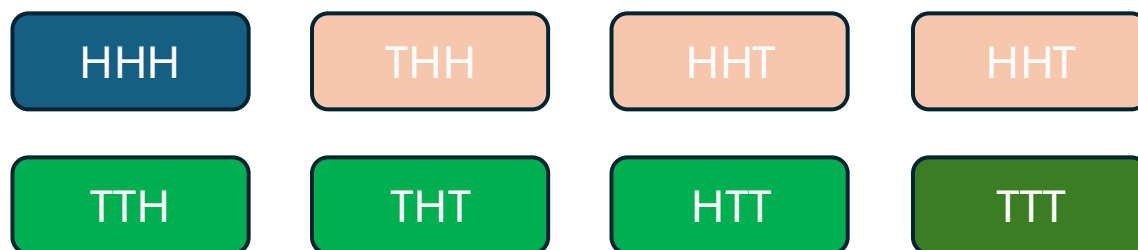
8 in total

In binomial distribution, we want to know the probability of a certain combination!

- How many combinations do we have?

What if we flip coin thrice?

- How many possible outcomes do we have?



- How many combinations do we have?

- 3 H 0 T: 1 permutation 
- 2 H 1 T: 3 permutation   
- 1 H 2 T: 3 permutation   
- 0 H 3 T: 1 permutation 

What's the probability for each combination?

- If we flip a fair coin thrice, what's the possibility of 2H 1T?
 - For each of the permutations, the probability is 0.125 ($=0.5*0.5*0.5$)
 - We have three permutations, so the probability is 0.375 ($=0.125*3$)
- What if we flip an unfair coin, with the head probability 0.6?
What's the probability of 2H 1T?
 - For each permutation: $0.6*0.6*(1-0.6) = 0.144$
 - Three permutations, so the probability is 0.432 ($=0.144*3$)

On to A Probability Distribution

The probability of k events, with probability p per event, in N trials is...

Number of ways to get the combination

Prob(k) = (permutations of N choose k) * probability of that combination

Probability of a combination = $(p)^k(1-p)^{(N-k)}$

Numbers of Permutations

- Two items: a, b
 - Permutations = ab, ba (2 total)
- Three items: a, b, c
 - Permutation = abc, acb, bac, bca, cab, cba (6 total)
- Four items: a, b, c, d
 - Permutations = abcd, abdc, acbd, acdb,... (24 total)
- # permutations = $N!$, where N is the number of different items
 - $N! = N \times (N-1) \times \cdots \times 2 \times 1$ ($N!$ reads N factorial)

Interchangeable Items

- In a binomial distribution swapping the position of the same items does not change the permutation!
 - Given the sequence of 6 coin flips = HTHHTH
 - I swap the first and last H, we still get HTHHTH
- There are 6 items, so $6! = 720$ sequences
 - but there are $4!$ changes in Hs, and $2!$ changes in Ts which do not change the permutation
- So we have $6!/(4! \times 2!)$ different permutations of 4 heads and 2 tails

“N choose k”

- This is a case of N choose k, where N is the length of sequence and k is the number of “successes”
 - Out of N objects we have k heads and (N-k) tails
 - Number of ways to get k option 1 and (N-k) option 2
- In our example (HTHHTH) N=6, k=4 head, N-k=2 tails
 - This generates $N!/(k! * (N-k)!)$ possible choices

Total # of sequences

of positions where item type 1 (heads)
can be swapped w/o change in combination

of positions where item type 2 (tails)
can be swapped w/o change in combination

The Binomial Distribution

Substituting our N choose k equation we get...

$$P(k \text{ outcomes}) = \{N!/(k! * (N-k)!)\} * p^k(1-p)^{(N-k)}$$

This is the *binomial probability distribution* function

A mathematical description of the probability of a getting k “successes” over N trials where the probability of success is p (and events are independent)

- Doing no math, what is 6 choose 6?
- HHHHHH
- 6 choose 6 gives us
- $6!/(6! \times 0!) \quad 0!=1$
- so this is
- $6!/6! = 1$
- In a sequence of 6 heads (successes) there is only a single permutation that fulfills that condition

Binomial Distribution in Python

- There are quite a few distributions built into Python **scipy** and they all have a common set of four functions

```
from scipy.stats import binom
```

- `binom.pmf`
 - **P**robability **M**ass **F**unction
- `binom.cdf`
 - **C**umulative **D**istribution **F**unction
- `binom.ppf`
 - **P**ercent **P**oint **F**unction
- `binom.rvs`
 - **R**andom **V**ariates



binom is the
abbreviation
for binomial
distribution

`binom.pmf()`

`binom.pmf` = tells you the probability of getting a specific number of successes in N trials with a specified probability per trial

What is the chance of getting exactly 50 heads out of 100 trials using a fair coin?

of successes, k

of trials, N

probability of success

`binom.pmf(50,100,0.5)`

`[1] 0.07958924`

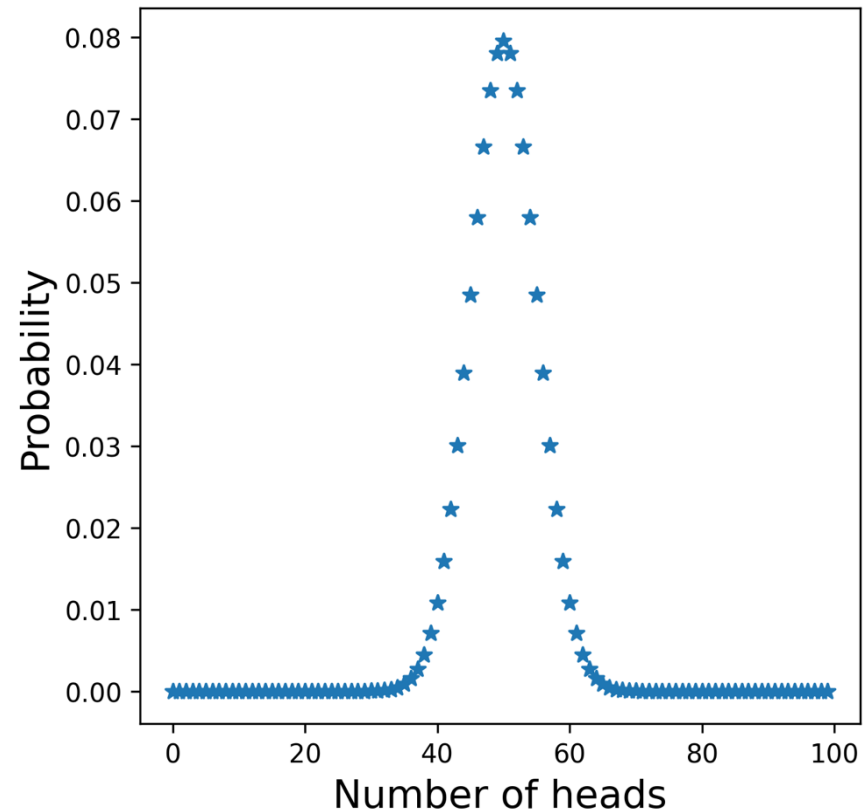
How about 51 heads?

`binom.pmf(51,100,0.5)`

`[1] 0.07802866`

binom.pmf of 50 flips of a fair coin

Most trials of 100 flips of a fair
coin look like they would be
between 40-60 heads



```
prob = []  
for x in range(0,100):  
    prob.append(binom.pmf(x,100,0.5))  
plt.figure(figsize=(10,5),dpi=300)  
plt.scatter(np.arange(0,100),prob, marker='*')  
plt.xlabel('Number of heads', fontsize=15)  
plt.ylabel('Probability', fontsize=15)
```

`binom.cdf()`

What if we wanted to know the chance of 50 heads or less?

Intuitively, it should be 50%

The chance of 50 or less would be $P(50)+P(49)+P(48)+\dots+P(0)$

`binom.cdf` gives the cumulative sum of at least k successes given N trials with probability p success per trial

`binom.cdf(50, 100, 0.5)`  parameters are again k , N , and p in that order
[1] 0.5397946

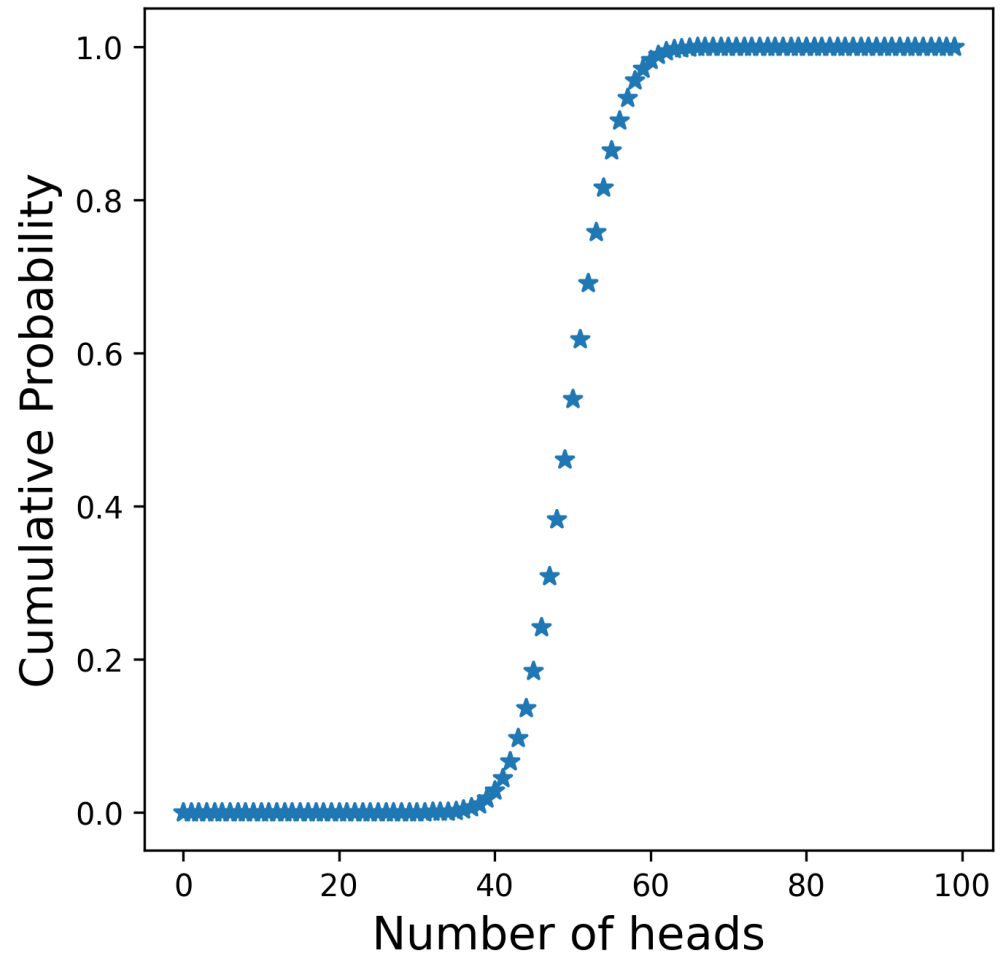
Why greater than 50%???

Actually 51 ways to get 50 or fewer heads (0, 1, 2, 3, ..., 49, 50)!

`binom.cdf` is giving us the probability of a given value or less, note the strong S shape here

What if you want the probability of greater than k successes?

You can do `1 - binom.cdf`



```
prob = []
for x in range(0,100):
    prob.append(binom.cdf(x,100,0.5))
plt.figure(figsize=(10,5),dpi=300)
plt.scatter(np.arange(0,100),prob, marker='*')
plt.xlabel('Number of heads', fontsize=15)
plt.ylabel('Cumulative Probability', fontsize=15)
```

binom.ppf

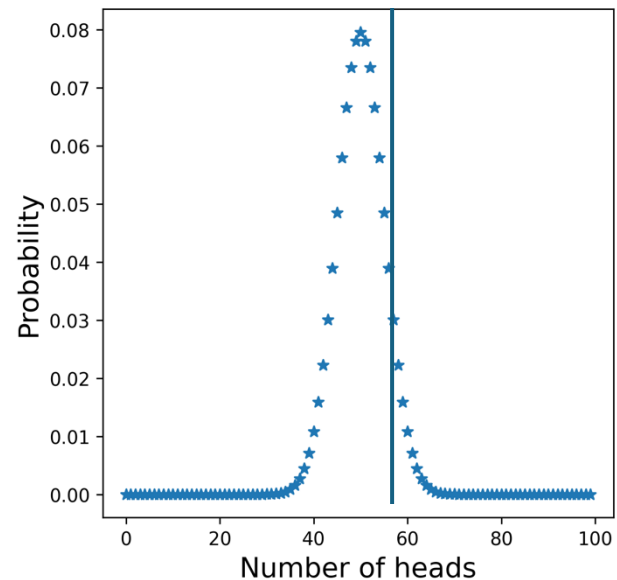
In 100 coin flips, most trials would yield ~50 heads

We might want to ask, where on this picture would we put the blue line, so that only 5% of the outcomes are to the right of it?

This is called a quantile function for the distribution

`binom.ppf(0.95, 100, 0.5)`
[1] 58

percentile of interest



Only 5% of the trials would have more than 58 heads

binom.rvs

Generates a random sequence of binomial values

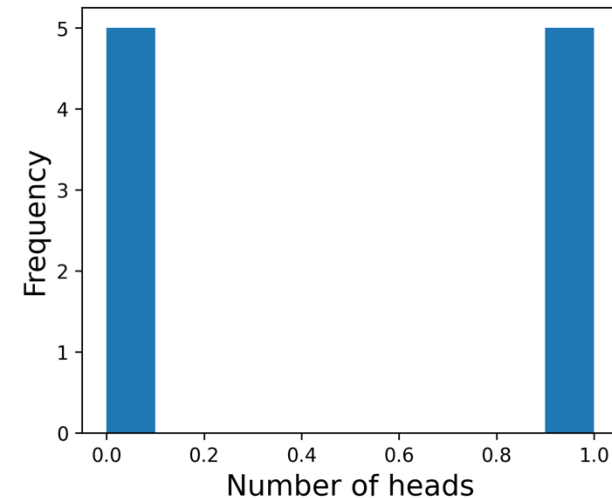
`binom.rvs(1, 0.5, size=10)`

length of sequence

```
[1] 0 0 1 0 0 1 0 1 1 1
```

This is a simulation of 10 flips of a fair coin

```
plt.figure(figsize=(5,2),dpi=300)
plt.hist(binom.rvs(1, 0.5, size=10))
plt.xlabel('Number of heads', fontsize=15)
plt.ylabel('Frequency', fontsize=15)
```



Probability Distribution Summary

- `binom.pmf()` returns the probability of getting exactly a certain number of success
 - 1- `binom.pmf` is the probability of any other possibility
- `binom.cdf()` returns the cumulative probability of getting less than or equal to a certain number of successes
 - 1- `binom.cdf` is the probability of getting more than a certain number of success
- `binom.ppf()` returns the number of successes associated with a particular percentile value
- `binom.rvs()` returns a set of numbers randomly drawn from a binomial distribution

Normal Distribution (mean, sd)

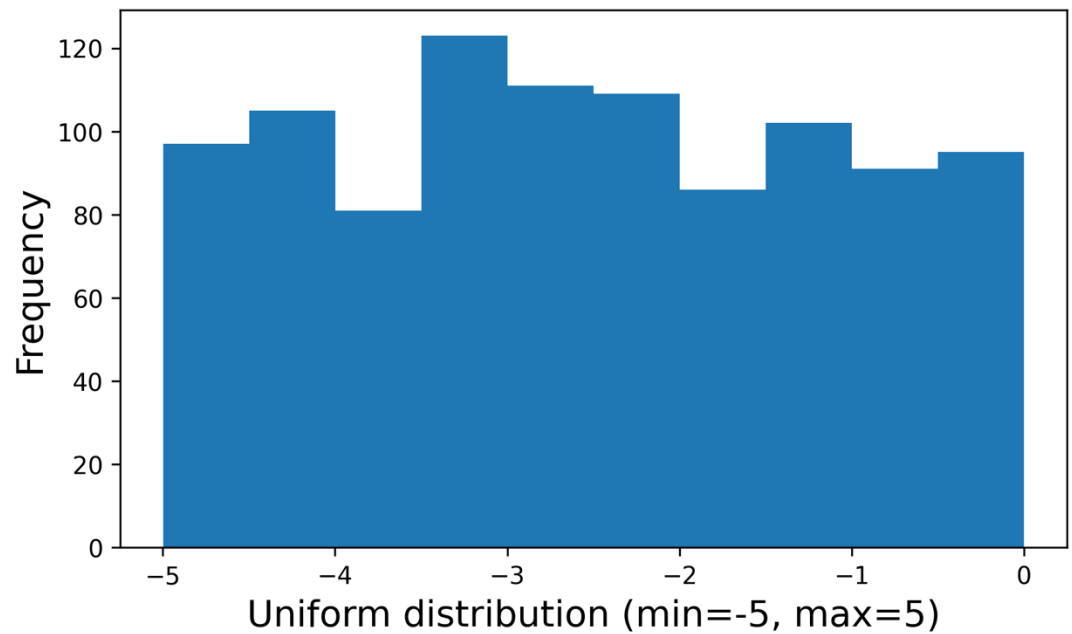
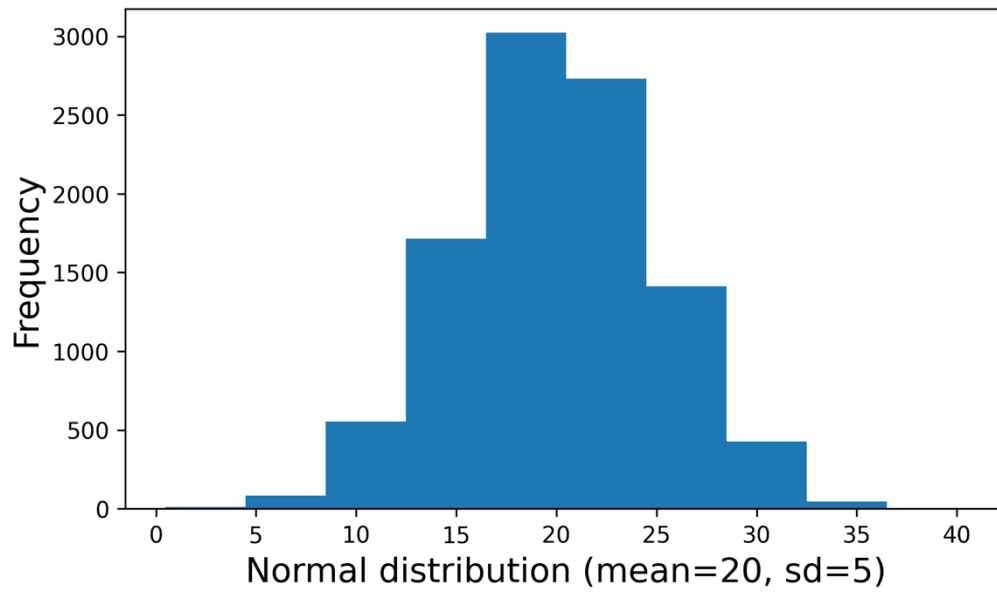
```
from scipy.stats import norm
```

- Normal distribution is your simple bell curve
 - `norm.pdf(20, 20, 5)` = probability of getting the value 20 from a normal distribution with mean=20, sd=5
 - `norm.cdf(20, 20, 5)` = cumulative probability of getting a value equal to or less than 20
 - `norm.ppf(0.5, 20, 5)` = returns value where 50% of the cumulative probability is below
 - `norm.rvs(20, 5, size=20)` = generates 20 random numbers drawn from a normal distribution (mean=20, sd=5)

Uniform Distribution (min, max)

```
from scipy.stats import uniform
```

- Uniform distribution is where all values have equal probability
 - `uniform.pdf(5, 1, 10)` = probability of getting the value 5 from a uniform distribution between 1 and 10
 - `uniform.cdf(5, 1, 10)` = cumulative probability of getting a value equal to or less than 5
 - `uniform.ppf(0.5, 1, 10)` = returns value where 50% of the cumulative probability is below
 - `uniform.rvs(1, 10, size=10)` = generates 20 random numbers drawn from a uniform (min=1, max=10)



Other Descriptors of Distributions

- skewness()
 - measures the asymmetry of a distribution/dataset

$$\tilde{\mu}_3 = \frac{\sum_i^N (X_i - \bar{X})^3}{(N - 1) * \sigma^3}$$

$\tilde{\mu}_3$ = skewness

N = number of variables in the distribution

X_i = random variable

$\bar{X}$ = mean of the distribution

σ = standard deviation

- kurtosis()
 - measure of the peakiness of a distribution/dataset

- Both in “scipy” package

$$\gamma_2 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^2}$$

```
from scipy.stats import skew, kurtosis
```

Skewness

- Value increases as asymmetry of around the mean increases

$$\tilde{\mu}_3 = \frac{\sum_i^N (X_i - \bar{X})^3}{(N - 1) * \sigma^3}$$

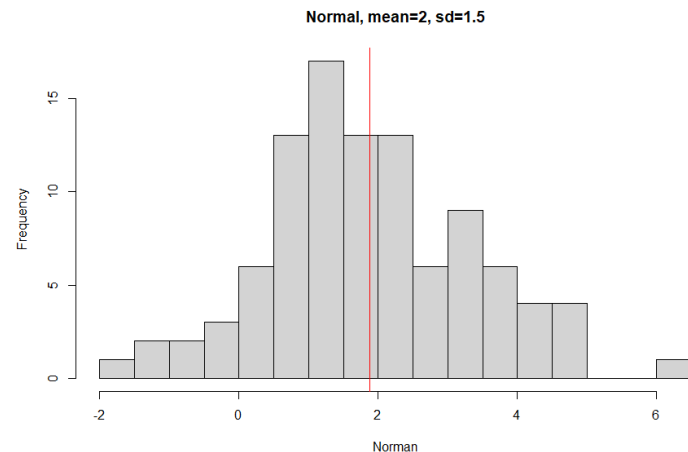
$\tilde{\mu}_3$ = skewness

N = number of variables in the distribution

X_i = random variable

$\bar{X}$ = mean of the distribution

σ = standard deviation



Skewness

- Value increases as asymmetry of around the mean increases
- Value decreases as standard deviation increases

$$\tilde{\mu}_3 = \frac{\sum_i^N (X_i - \bar{X})^3}{(N - 1) * \sigma^3}$$

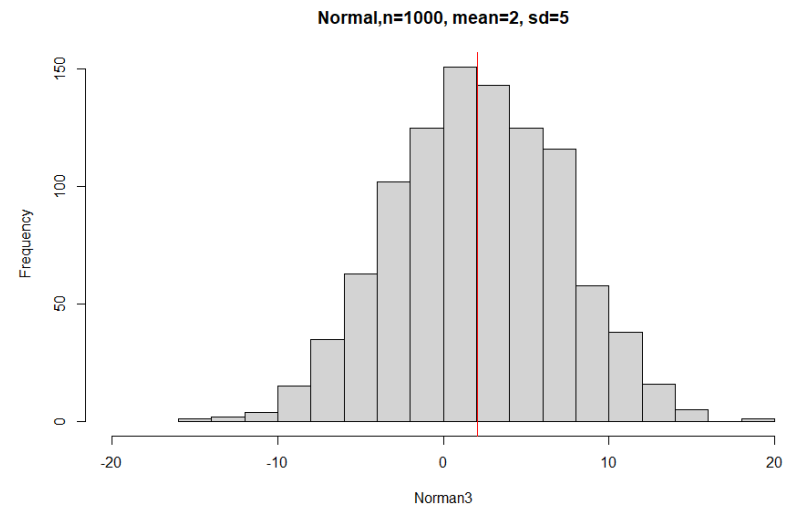
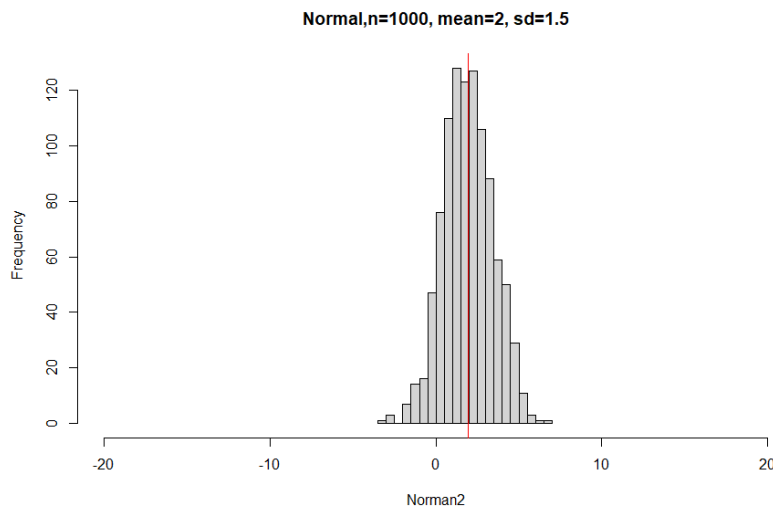
$\tilde{\mu}_3$ = skewness

N = number of variables in the distribution

X_i = random variable

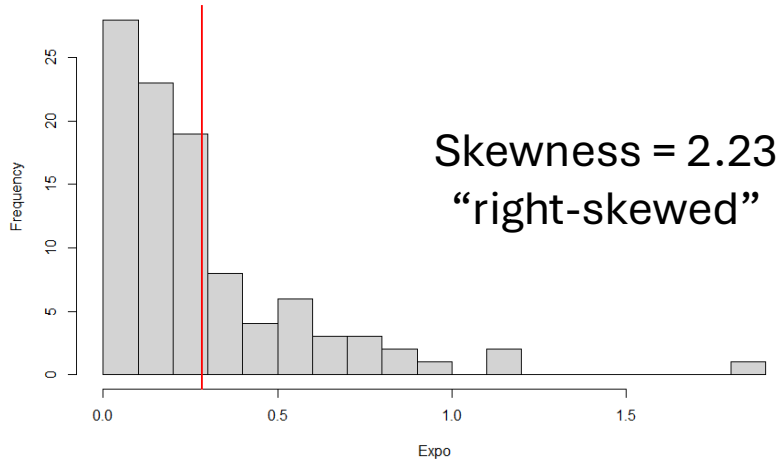
$\bar{X}$ = mean of the distribution

σ = standard deviation

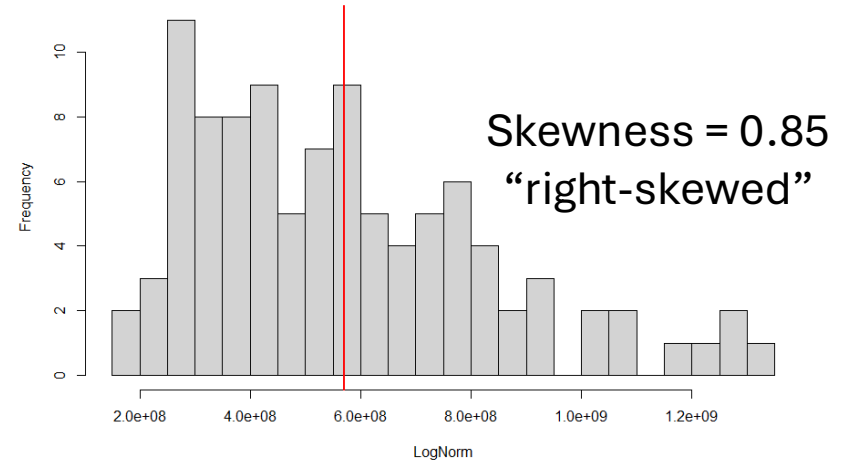


Skewness

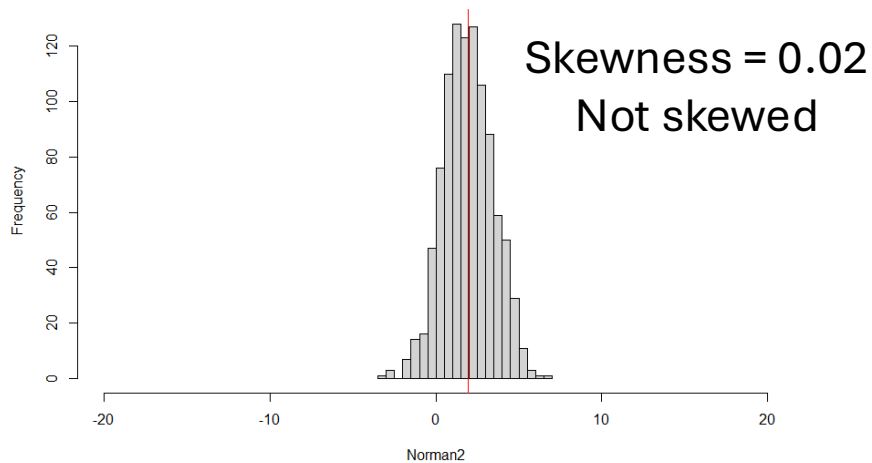
Exponential



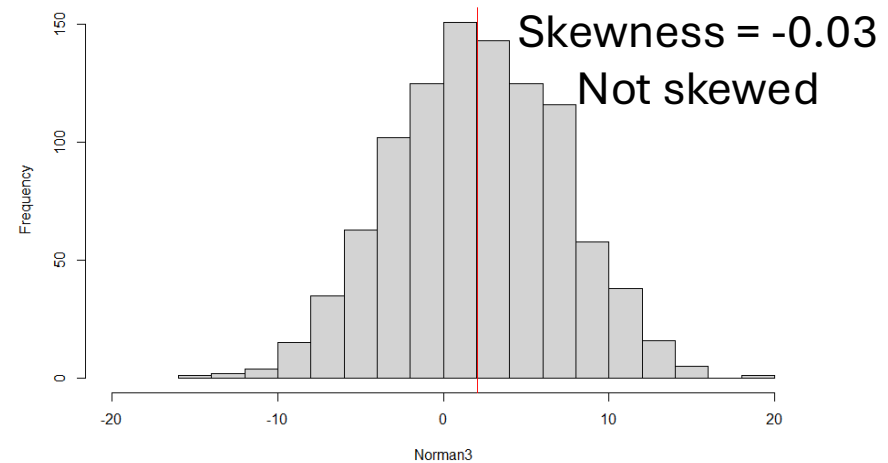
LogNormal



Normal,n=1000, mean=2, sd=1.5



Normal,n=1000, mean=2, sd=5



Kurtosis

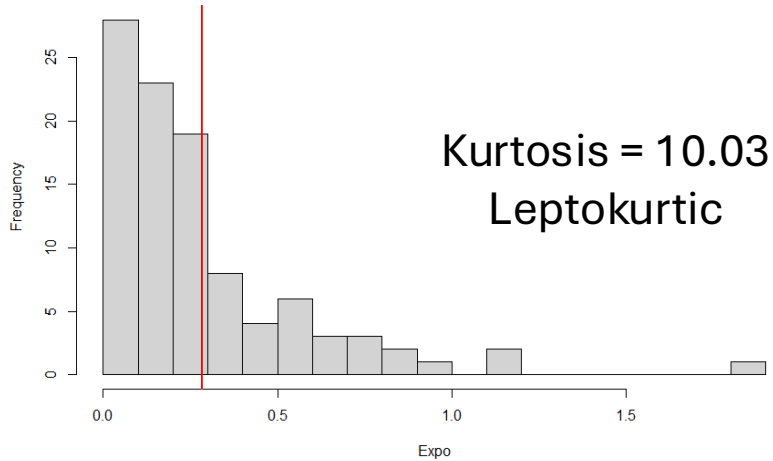
- Value increases as values are more clustered around the mean value

$$\gamma_2 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^2}$$

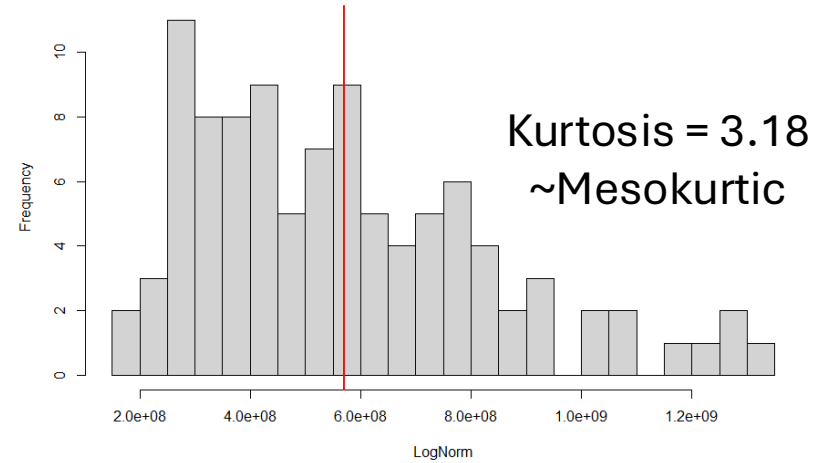
- Value decreases as sample size grows, all else being equal
 - More samples = more likely to find extreme values of a distribution
- Platykurtic (kurtosis $\ll 3$, small tails compared to normal)
- Mesokurtic (kurtosis ≈ 3 , within range of normal distribution)
- Leptokurtic (kurtosis $\gg 3$, large tails compared to normal)

Kurtosis

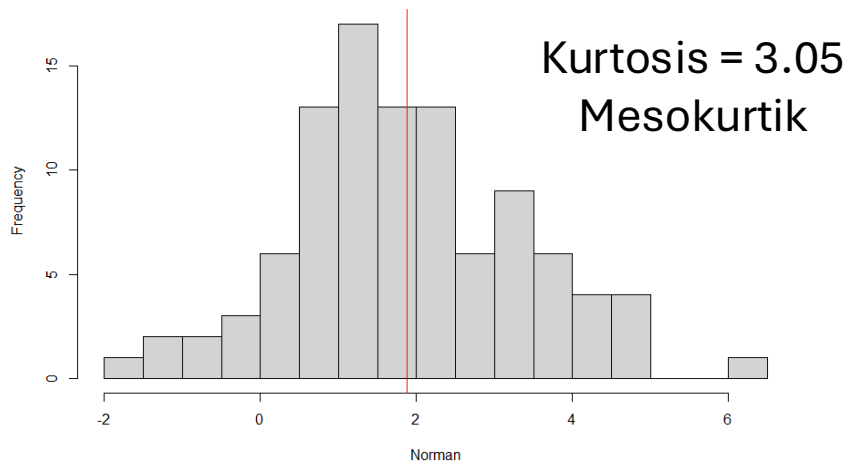
Exponential



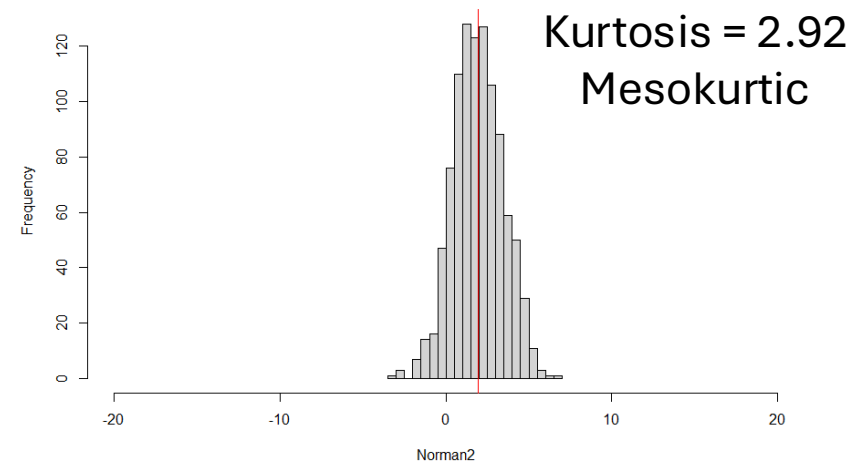
LogNormal



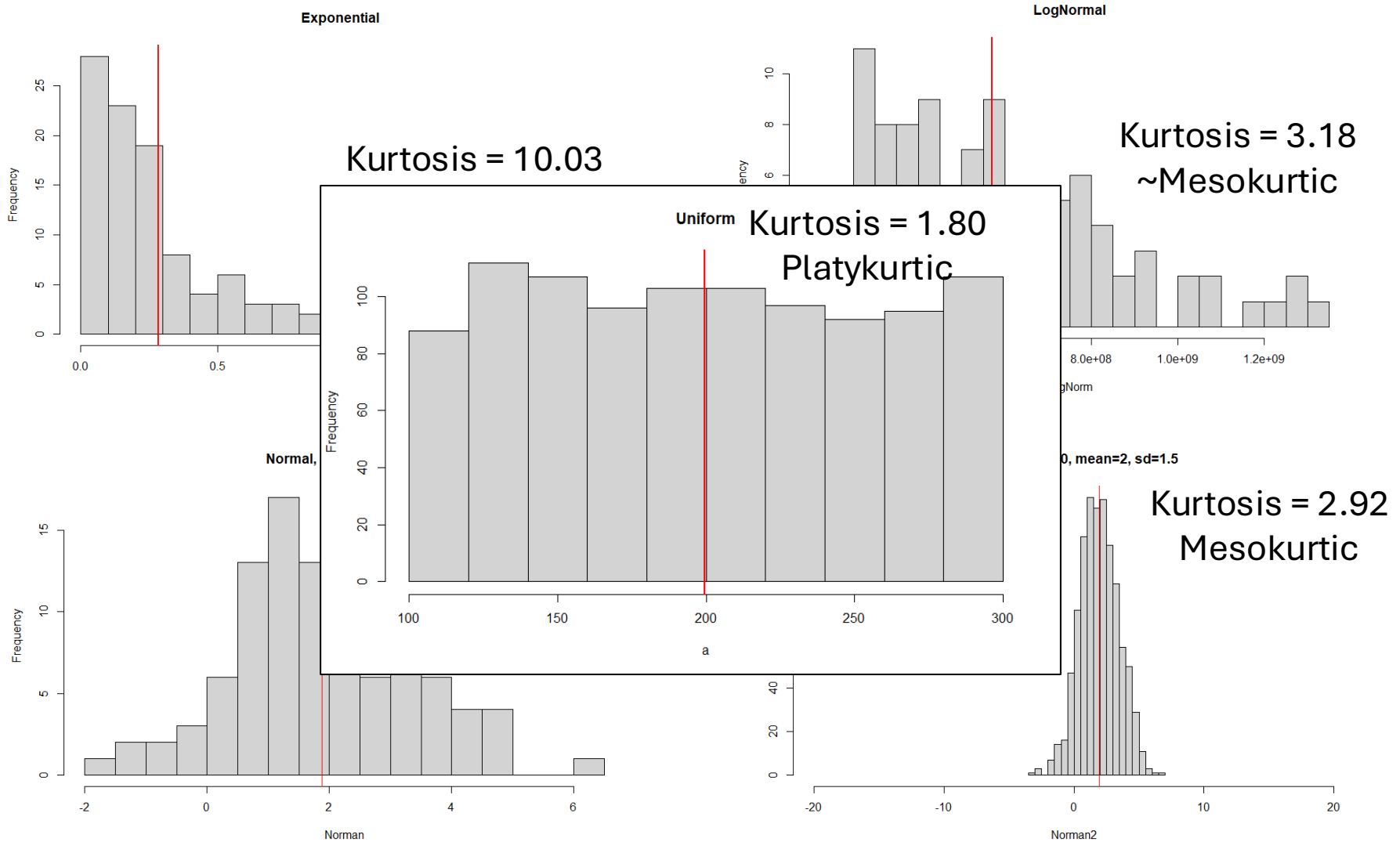
Normal, mean=2, sd=1.5



Normal, n=1000, mean=2, sd=1.5



Kurtosis



Tests of Distribution Fit

- One of the most common early steps in analyses is to see whether a distribution fits the data
- Check for normality to see if other tests can be used
 - ex. t-test
- Shapiro-Wilk
- Kolmogorov-Smirnov

Shapiro-Wilk Normality Test

- Tests whether a set of data are normally distributed
- Relatively weak test, especially at small sample sizes because the normal distribution is a good first approximation of most distributions

- `from scipy.stats import shapiro`
- `shapiro(x)`
 - `stat, p = shapiro(x)`
 - `print("Test statistic:", stat)`
 - `print("p-value:", p)`

Kolmogorov-Smirnov tests

- Tests whether one distribution is consistent with another
- Again, not particularly strong with sparse data
 - `from scipy.stats import kstest`

```
>>> x = stats.norm.rvs(size=100, random_state=rng)
>>> stats.kstest(x, stats.norm.cdf)
```

Output:

```
KstestResult(statistic=0.05345882212970396,
             pvalue=0.9227159037744717,
             statistic_location=-1.2451343873745018,
             statistic_sign=1)
```

Kolmogorov-Smirnov tests

- Tests whether one distribution is consistent with another
- Again, not particularly strong with sparse data
 - `from scipy.stats import kstest`

```
>>> sample1 = stats.laplace.rvs(size=105, random_state=rng)
>>> sample2 = stats.laplace.rvs(size=95, random_state=rng)
>>> stats.kstest(sample1, sample2)
```

Output:

```
KstestResult(statistic=0.11779448621553884,
             pvalue=0.4494256912629795,
             statistic_location=0.6138814275424155,
             statistic_sign=1)
```

Today's Learning Outcomes

1. Be able to explain the binomial and normal distributions
2. Know pmf/pdf , cdf , ppf , and rvs are and do
3. Know how to test for normality of a dataset